For each real number t, identify a point $B(x, y)$ on the unit circle such that the arc length AB is equal to t. If t is positive, choose the point $B(x, y)$ in the anticlockwise direction, otherwise choose it in the clockwise direction. Let θ be the angle subtended by the arc AB at the centre. In this way, we have a function $\mathrm{w}(t)$ associating a real number t to a point on the unit circle. Such a function is called a **wrapping function** . Then $s = r\theta$ gives arc length $t = \theta$.

Now, define $\sin t = \sin\theta$ and $\cos t = \cos\theta$.

Clearly, $\sin t = \sin\theta = y$ and $\cos t = \cos\theta = x$.

Using $\sin t$ and $\cos t$, other trigonometric functions can be defined as functions of real numbers.

(i) $B(x, y) = B(\cos t, \sin t)$ is a point on the unit circle.
Thus, $-1 \leq \cos t \leq 1$ and $-1 \leq \sin t \leq 1$ for any real number t.

(ii) Wrapping function $\mathrm{w}(t)$ is analogous to wrapping a line around a circle.

(iii) The value of a trigonometric function of a real number t is its value at the angle t radians.

(iv) Trigonometric functions of real numbers are used to model phenomena like waves, oscillations, that occur in regular intervals.

Example 3.8 The terminal side of an angle θ in standard position passes through the point $(3, -4)$. Find the six trigonometric function values at an angle θ.

Solution:

Let $B(x, y) = B(3, -4)$, OA be the initial side and OB be the terminal side of the angle θ in the standard position.

Then $\angle AOB$ is the angle θ and θ lies in the IV quadrant. Also,

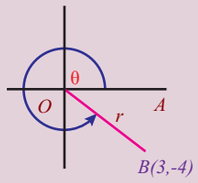


Figure 3.9

$$OB = r,\ r = \sqrt{x^2 + y^2} = \sqrt{3^2 + (-4)^2} = 5$$

$$x = 3,\ \ y = -4 \ \text{ and } \ r = 5, \ \text{ we have}$$

$$\sin\theta = \frac{y}{r} = -\frac{4}{5};\quad \cos\theta = \frac{x}{r} = \frac{3}{5};\quad \tan\theta = \frac{y}{x} = -\frac{4}{3};$$

$$\operatorname{cosec}\theta = \frac{r}{y} = -\frac{5}{4};\quad \sec\theta = \frac{r}{x} = \frac{5}{3};\quad \cot\theta = \frac{x}{y} = -\frac{3}{4}.$$

Signs of Trigonometric functions

Consider a unit circle with centre at the origin. Let θ be in standard position. Let $P(x, y)$ be the point on the unit circle corresponding to the angle θ. Then, $\cos\theta = x$, $\sin\theta = y$ and $\tan\theta = \frac{y}{x}$. The values of x and y are positive or negative depending on the quadrant in which P lies.

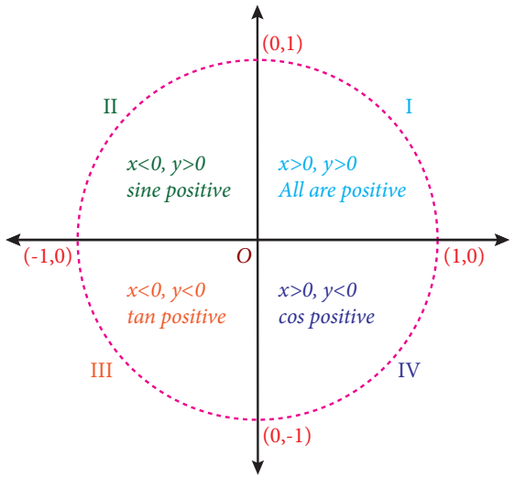


Figure 3.10

In the first quadrant:

$\cos\theta = x > 0$ (positive); $\sin\theta = y > 0$ (positive)

Thus, $\cos\theta$ and $\sin\theta$ and hence all trigonometric functions are positive in the first quadrant.

In the second quadrant:

$\cos\theta = x < 0$ (negative); $\sin\theta = y > 0$ (positive)